DBMS Lab Assignment 8 - Solution

Set operators (UNION, UNION ALL, INTERSECT, MINUS) on Sports_Stud and Cultural_Stud. Each question shows the Oracle SQL followed by its output.

*Note: outputs were produced by running equivalent statements on a SQLite database with the same tables and data (MINUS is EXCEPT in SQLite), formatted like SQL*Plus. SYSDATE is taken as 09-Nov-2026.*

1. Create the tables (constraints named)

SQL

```
CREATE TABLE Sports_Stud (
  ROLL       NUMBER(3)    CONSTRAINT pk_sports PRIMARY KEY,
  NAME       VARCHAR2(20) CONSTRAINT uq_sports_name UNIQUE,
  DEPT       VARCHAR2(2)  CONSTRAINT chk_sports_dept CHECK (DEPT IN ('IT','CS')),
  SPORTS_FEE NUMBER(6)    CONSTRAINT chk_sports_fee CHECK (SPORTS_FEE >= 1000));

CREATE TABLE Cultural_Stud (
  ROLL     NUMBER(3)    CONSTRAINT pk_cul PRIMARY KEY,
  NAME     VARCHAR2(20) CONSTRAINT uq_cul_name UNIQUE,
  DEPT     VARCHAR2(2)  CONSTRAINT chk_cul_dept CHECK (DEPT IN ('IT','CS')),
  CUL_FEES NUMBER(6)    CONSTRAINT chk_cul_fee CHECK (CUL_FEES >= 600));

INSERT INTO Sports_Stud VALUES (1,'Aman','IT',1200);
INSERT INTO Sports_Stud VALUES (7,'Anurag','CS',1500);
INSERT INTO Sports_Stud VALUES (2,'Binay','IT',1000);
INSERT INTO Sports_Stud VALUES (8,'Chiranjit','CS',1700);
INSERT INTO Cultural_Stud VALUES (1,'Aman','IT',700);
INSERT INTO Cultural_Stud VALUES (2,'Binay','IT',700);
INSERT INTO Cultural_Stud VALUES (3,'Bipul','CS',600);
INSERT INTO Cultural_Stud VALUES (4,'Joy','CS',600);
INSERT INTO Cultural_Stud VALUES (5,'Surojit','IT',700);
INSERT INTO Cultural_Stud VALUES (6,'Wasim','CS',600);
COMMIT;
```

OUTPUT

```
Table created.

Table created.

4 rows created. (Sports_Stud)
6 rows created. (Cultural_Stud)

ROLL  NAME       DEPT  SPORTS_FEE
----  ---------  ----  ----------
   1  Aman       IT          1200
   2  Binay      IT          1000
   7  Anurag     CS          1500
   8  Chiranjit  CS          1700

4 rows selected.

ROLL  NAME     DEPT  CUL_FEES
----  -------  ----  --------
   1  Aman     IT         700
   2  Binay    IT         700
   3  Bipul    CS         600
   4  Joy      CS         600
   5  Surojit  IT         700
   6  Wasim    CS         600

6 rows selected.
```

Note: The sheet's 'Cultural_Stud should be >= 600' is read as CUL_FEES >= 600. NAME is unique in each table.

2. All distinct students in the organization (both departments)

SQL

```
SELECT roll, name, dept FROM Sports_Stud
UNION
SELECT roll, name, dept FROM Cultural_Stud;
```

OUTPUT

```
ROLL  NAME       DEPT
----  ---------  ----
   1  Aman       IT
   2  Binay      IT
   3  Bipul      CS
   4  Joy        CS
   5  Surojit    IT
   6  Wasim      CS
   7  Anurag     CS
   8  Chiranjit  CS

8 rows selected.
```

Note: UNION removes the duplicates (Aman and Binay are in both tables).

3. All students, duplicates included

SQL

```
SELECT roll, name, dept FROM Sports_Stud
UNION ALL
SELECT roll, name, dept FROM Cultural_Stud;
```

OUTPUT

```
ROLL  NAME       DEPT
----  ---------  ----
   1  Aman       IT
   2  Binay      IT
   7  Anurag     CS
   8  Chiranjit  CS
   1  Aman       IT
   2  Binay      IT
   3  Bipul      CS
   4  Joy        CS
   5  Surojit    IT
   6  Wasim      CS

10 rows selected.
```

4. Students in both the cultural and sports groups

SQL

```
SELECT roll, name, dept FROM Sports_Stud
INTERSECT
SELECT roll, name, dept FROM Cultural_Stud;
```

OUTPUT

```
ROLL  NAME   DEPT
----  -----  ----
   1  Aman   IT
   2  Binay  IT

2 rows selected.
```

5. Students in only the sports group

SQL

```
SELECT roll, name, dept FROM Sports_Stud
MINUS
SELECT roll, name, dept FROM Cultural_Stud;
```

OUTPUT

```
ROLL  NAME       DEPT
----  ---------  ----
   7  Anurag     CS
   8  Chiranjit  CS

2 rows selected.
```

6. Students in only the cultural group

SQL

```
SELECT roll, name, dept FROM Cultural_Stud
MINUS
SELECT roll, name, dept FROM Sports_Stud;
```

OUTPUT

```
ROLL  NAME     DEPT
----  -------  ----
   3  Bipul    CS
   4  Joy      CS
   5  Surojit  IT
   6  Wasim    CS

4 rows selected.
```

7. Roll, sports fee and cultural fee of students in both groups

SQL

```
SELECT s.roll, s.sports_fee, c.cul_fees
FROM   Sports_Stud s JOIN Cultural_Stud c ON s.roll = c.roll;
```

OUTPUT

```
ROLL  SPORTS_FEE  CUL_FEES
----  ----------  --------
   1        1200       700
   2        1000       700

2 rows selected.
```

Note: A set operator cannot put two fee columns side by side, so a join on ROLL is used (INTERSECT on roll would only identify the students).

8. Add a BIRTHDAY field to both tables and fill it in

SQL

```
ALTER TABLE Sports_Stud   ADD (birthday DATE);
ALTER TABLE Cultural_Stud ADD (birthday DATE);

UPDATE Sports_Stud SET birthday = DATE '2004-03-15' WHERE name = 'Aman';
UPDATE Sports_Stud SET birthday = DATE '2003-11-05' WHERE name = 'Binay';
UPDATE Sports_Stud SET birthday = DATE '2004-02-29' WHERE name = 'Anurag';
UPDATE Sports_Stud SET birthday = DATE '2005-05-18' WHERE name = 'Chiranjit';
UPDATE Cultural_Stud SET birthday = DATE '2004-03-15' WHERE name = 'Aman';
UPDATE Cultural_Stud SET birthday = DATE '2003-11-05' WHERE name = 'Binay';
UPDATE Cultural_Stud SET birthday = DATE '2004-07-22' WHERE name = 'Bipul';
UPDATE Cultural_Stud SET birthday = DATE '2005-01-30' WHERE name = 'Joy';
UPDATE Cultural_Stud SET birthday = DATE '2003-12-12' WHERE name = 'Surojit';
UPDATE Cultural_Stud SET birthday = DATE '2004-09-09' WHERE name = 'Wasim';
COMMIT;

DESC Sports_Stud;
DESC Cultural_Stud;
SELECT * FROM Sports_Stud;
SELECT * FROM Cultural_Stud;
```

OUTPUT

```
Table altered.

Table altered.

10 rows updated.

SQL> DESC Sports_Stud
Name         Null?     Type
-----------  --------  ------------
ROLL         NOT NULL  NUMBER(3)
NAME                   VARCHAR2(20)
DEPT                   VARCHAR2(2)
SPORTS_FEE             NUMBER(6)
BIRTHDAY               DATE

SQL> DESC Cultural_Stud
Name       Null?     Type
---------  --------  ------------
ROLL       NOT NULL  NUMBER(3)
NAME                 VARCHAR2(20)
DEPT                 VARCHAR2(2)
CUL_FEES             NUMBER(6)
BIRTHDAY             DATE

ROLL  NAME       DEPT  SPORTS_FEE  BIRTHDAY
----  ---------  ----  ----------  ---------
   1  Aman       IT          1200  15-MAR-04
   2  Binay      IT          1000  05-NOV-03
   7  Anurag     CS          1500  29-FEB-04
   8  Chiranjit  CS          1700  18-MAY-05

4 rows selected.

ROLL  NAME     DEPT  CUL_FEES  BIRTHDAY
----  -------  ----  --------  ---------
   1  Aman     IT         700  15-MAR-04
   2  Binay    IT         700  05-NOV-03
   3  Bipul    CS         600  22-JUL-04
   4  Joy      CS         600  30-JAN-05
   5  Surojit  IT         700  12-DEC-03
   6  Wasim    CS         600  09-SEP-04

6 rows selected.
```

Note: Birthdays are my own choice. Aman and Binay appear in both tables, so they get the same date in each.

9. Name, age and month of birth of all distinct students

SQL

```
SELECT name,
       FLOOR(MONTHS_BETWEEN(SYSDATE, birthday) / 12) AS age,
       TO_CHAR(birthday, 'Month') AS birth_month
FROM  (SELECT name, birthday FROM Sports_Stud
       UNION
       SELECT name, birthday FROM Cultural_Stud);
```

OUTPUT

```
NAME       AGE  BIRTH_MONTH
---------  ---  -----------
Aman        22  March
Anurag      22  February
Binay       23  November
Bipul       22  July
Chiranjit   21  May
Joy         21  January
Surojit     22  December
Wasim       22  September

8 rows selected.
```

Note: Ages are as of SYSDATE = 09-Nov-2026.

10. Last day of the month in which each student was born (using a set operator)

SQL

```
SELECT name, birthday, LAST_DAY(birthday) AS last_day_of_month
FROM   Sports_Stud
UNION
SELECT name, birthday, LAST_DAY(birthday)
FROM   Cultural_Stud;
```

OUTPUT

```
NAME       BIRTHDAY   LAST_DAY_OF_MONTH
---------  ---------  -----------------
Aman       15-MAR-04  31-MAR-04
Anurag     29-FEB-04  29-FEB-04
Binay      05-NOV-03  30-NOV-03
Bipul      22-JUL-04  31-JUL-04
Chiranjit  18-MAY-05  31-MAY-05
Joy        30-JAN-05  31-JAN-05
Surojit    12-DEC-03  31-DEC-03
Wasim      09-SEP-04  30-SEP-04

8 rows selected.
```

Note: UNION covers every student once; Anurag's last day is 29-Feb-04 because 2004 is a leap year.

11. Sports_Stud details in sentence format

SQL

```
SELECT name || ' who studies in ' || dept || ' dept pays Rs' ||
       sports_fee || ' for sports.' AS details
FROM   Sports_Stud;
```

OUTPUT

```
DETAILS
-----------------------------------------------------
Aman who studies in IT dept pays Rs1200 for sports.
Binay who studies in IT dept pays Rs1000 for sports.
Anurag who studies in CS dept pays Rs1500 for sports.
Chiranjit who studies in CS dept pays Rs1700 for sports.

4 rows selected.
```

12. IT students, and the total money earned from sports and from cultural

SQL

```
-- details of all IT students
SELECT roll, name, dept FROM Sports_Stud   WHERE dept = 'IT'
UNION
SELECT roll, name, dept FROM Cultural_Stud WHERE dept = 'IT';

-- total money earned from each group, and overall
SELECT (SELECT SUM(sports_fee) FROM Sports_Stud)   AS total_sports,
       (SELECT SUM(cul_fees)   FROM Cultural_Stud) AS total_cultural,
       (SELECT SUM(sports_fee) FROM Sports_Stud) +
       (SELECT SUM(cul_fees)   FROM Cultural_Stud) AS grand_total
FROM   dual;
```

OUTPUT

```
ROLL  NAME     DEPT
----  -------  ----
   1  Aman     IT
   2  Binay    IT
   5  Surojit  IT

3 rows selected.

TOTAL_SPORTS  TOTAL_CULTURAL  GRAND_TOTAL
------------  --------------  -----------
        5400            3900         9300

1 row selected.
```

Note: Sports total = 1200+1500+1000+1700 = 5400; cultural total = 700+700+600+600+700+600 = 3900.

13. Names of all students in descending order

SQL

```
SELECT name FROM Sports_Stud
UNION
SELECT name FROM Cultural_Stud
ORDER BY name DESC;
```

OUTPUT

```
NAME
---------
Wasim
Surojit
Joy
Chiranjit
Bipul
Binay
Anurag
Aman

8 rows selected.
```

Note: UNION lists each student once. Use UNION ALL if you want Aman and Binay twice.

14. Drop all the domain constraints from both tables

SQL

```
ALTER TABLE Sports_Stud   DROP CONSTRAINT chk_sports_dept;
ALTER TABLE Sports_Stud   DROP CONSTRAINT chk_sports_fee;
ALTER TABLE Cultural_Stud DROP CONSTRAINT chk_cul_dept;
ALTER TABLE Cultural_Stud DROP CONSTRAINT chk_cul_fee;

SELECT table_name, constraint_name, constraint_type
FROM   user_constraints
WHERE  table_name IN ('SPORTS_STUD','CULTURAL_STUD');
```

OUTPUT

```
Table altered.

Table altered.

Table altered.

Table altered.

TABLE_NAME     CONSTRAINT_NAME  C
-------------  ---------------  -
CULTURAL_STUD  PK_CUL           P
CULTURAL_STUD  UQ_CUL_NAME      U
SPORTS_STUD    PK_SPORTS        P
SPORTS_STUD    UQ_SPORTS_NAME   U

4 rows selected.
```

Note: Only the primary-key and unique constraints remain (the four CHECK constraints are gone).

15. Maximum, minimum and average sports fee

SQL

```
SELECT MAX(sports_fee) AS max_fee,
       MIN(sports_fee) AS min_fee,
       AVG(sports_fee) AS avg_fee
FROM   Sports_Stud;
```

OUTPUT

```
MAX_FEE  MIN_FEE  AVG_FEE
-------  -------  -------
   1700     1000     1350

1 row selected.
```